Data Mining Project

BikeReg: Predicting Urban Bicycle Flow in Mannheim

Submitted by Team 10
Romeo Türemis (1696319) Ole Behre (1832323)
Sarah Suleiman (1837321) Nishit Suthar (2189914)
Manuel Wirth (2266428)

Code available at: https://github.com/OleBehre/BikeReg

Submitted to Data and Web Science Group
Prof. Dr. Sven Hertling, University of Mannheim

Contents

1 Application area and goals

Mannheim, the birthplace of the bicycle invented by Karl Drais in 1817, is continuing this legacy by strengthening its identity as a Cycling City (Stadt Mannheim a). A central part of this transition is creating traffic infrastructure that responds more effectively to the needs of cyclists.

This project focuses on Urban Traffic Management and short-term forecasting. Currently, many traffic lights in Mannheim follow a fixed timing or remain oriented toward car traffic, which often forces cyclists to stop unnecessarily. To improve flow, the city could explore concepts such as real-time "Green Waves," inspired by pilots in Amsterdam and Copenhagen (Centre for Public Impact), where signals adjust to approaching groups of riders.

The goal of *BikeReg* is to forecast bicycle traffic on important bottlenecks such as the Kurpfalzbrücke with a one-hour prediction horizon. Reliable forecasts of intense bike traffic allow the city to take dynamic measures. These measures include temporarily allocating an extra lane for cyclists or giving them priority at traffic lights, which helps maintain smooth and safe movement across the bridge.

Because quick changes in demand are common, the model must detect short-term fluctuations and react to emerging trends. The main data-mining objective is to minimize the Mean Absolute Error (MAE) of hourly predictions. Accurate forecasts provide the basis for timely interventions that improve travel times for cyclists and strengthen sustainable urban mobility.

2 Dataset Profile

2.1 Data Sources and Scope

The city of Mannheim provides several datasets through it's Open Data-Portal (Stadt Mannheim b) including the 19 Eco-Counters for bike traffic. Each counter provides hourly data on the total number of bikes that drove past it measured using an induction loop in the ground. The dataset consists of a timestamp for each hourly entry, the total count that was measured, and redundant location information from the counter. This analysis focuses on the two counters located at the Kurpfalzbrücke, one for each side of the bridge. We chose these two counters because they consistently reported the highest readings for bike traffic (Stadt Mannheim c).

We included hourly weather data from the Deutscher Wetterdienst (DWD) as we assume that the weather has a significant impact on bike traffic. The DWD provides historical weather data through it's API (Deutscher Wetterdienst). Using the API, temperature means and precipitation height were aggregated for each timestamp of the Eco-Counter data.

A further external data source used in this analysis concerns public holidays in Baden-Württemberg, as public holidays are expected to influence commute volume. The holiday information was retrieved via the Open Holiday API (Open Holiday API) and merged with the existing dataset. The resulting binary holiday indicator was added as an additional feature.

Information on school holidays in Baden-Württemberg was also incorporated, as periods without regular schooling can significantly impact daily mobility patterns. The data was manually extracted (Schulferien.org) and mapped to the corresponding dates in the dataset. A binary feature was created to indicate whether a given day fell within a school holiday period.

Two additional features were manually aggregated: one binary indicator if a given day is during the lecture time of the University of Mannheim and another binary indicator if a public transit strike occurred. The strike data is likely incomplete as there are multiple sources (RNV, verdi, etc.) and inconsistent information for past strikes.

The full Eco-Counter dataset contains about 80,200 records, ranging from October 4, 2016 until now.

2.2 Exploratory Data Analysis

We conducted an Exploratory Data Analysis (EDA) to gain insights into the dataset and it's characteristics like trends and anomalies.

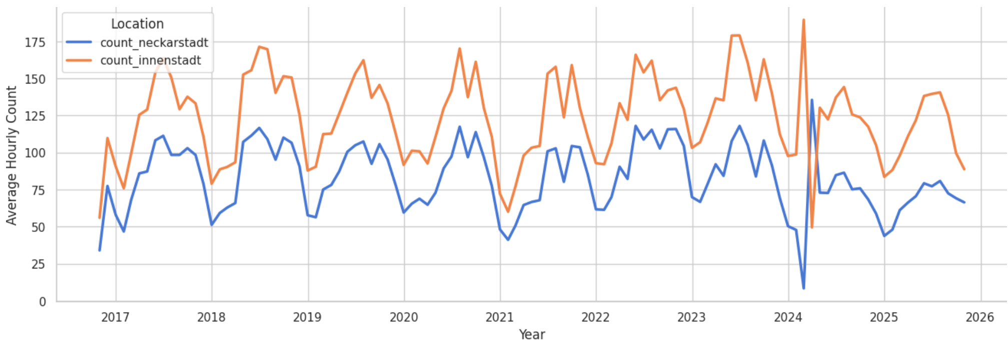

Figure 2.1: Average hourly bike counts aggregated monthly, showing seasonal trends.

Figure 2.1 shows the monthly aggregated counts of both Eco-Counters throughout the years. The figure reveals clear seasonal patterns, with bike usage peaking consistently during the warmer months and dropping in winter. While both counters follow similar seasonal cycles, the counter for the Innenstadt side shows noticeably higher average counts across all years. The most noticeable anomaly occurs in early 2024, where one counter shows a sharp increase while the other simultaneously drops. This pattern is best explained by the major construction work on the bridge in March 2024: first, one lane was closed, forcing all bike traffic onto the opposite lane, and later the situation was reversed when the lanes switched. After the construction period, both counters record lower values compared to previous years. This sudden shift poses challenges for modeling, as it cannot be reliably captured by models trained solely on pre-2024 data.

Figure 2.2 reveals clear structural differences in hourly bicycle activity between weekdays and weekends. Weekdays show pronounced morning and evening peaks characteristic of commuter traffic, whereas weekends follow a flatter pattern with a later, broader midday peak driven by leisure mobility. The comparison highlights the strong influence of daily rhythm on bike usage, which later motivates the inclusion of temporal features such as `hour_of_day` and `is_weekend` in the modeling pipeline. It also underscores why lagged and cyclic patterns are expected to play an important role in prediction.

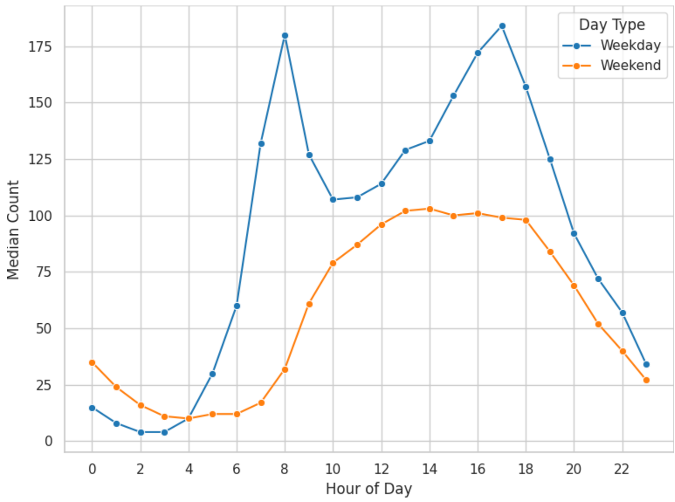

Figure 2.2: Hourly rhythm Weekday vs. Weekend.

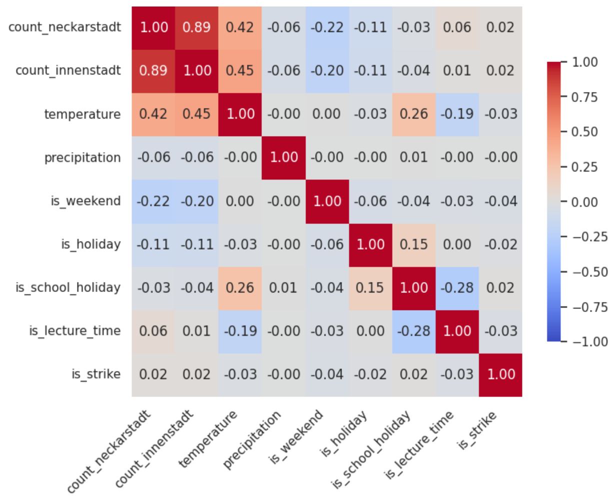

Figure 2.3: Correlation matrix of basic features.

Figure 2.3 presents the correlation matrix of the dataset's features. As expected, the two Eco-Counters exhibit a very strong positive correlation, indicating that both sides of the bridge respond to the same underlying traffic dynamics. Temperature emerges as the feature with the strongest positive association to bicycle counts, aligning with the intuition that warmer conditions encourage cycling. In contrast, `is_weekend` shows a negative relationship with bike volume, reflecting reduced commuter traffic on Saturdays and Sundays. Additional features such as holidays and precipitation show weaker correlations, suggesting that while they influence individual days, their effects are less dominant than daily cycles and weather.

Together, these findings highlight the central role of temporal patterns and environmental factors in shaping bicycle traffic and underscore and provide an insightful basis for preprocessing the dataset for our specifc requirements.

3 Preprocessing

We implemented a preprocessing pipeline to transform the raw data into a clean, feature-rich dataset suitable for our model training.

3.1 Data Cleaning and Validity Checks

The raw data contained several quality issues, including missing values, duplicate timestamps due to Daylight Saving Time (DST), and structural breaks caused by construction work. We addressed these issues through the following steps:

- **Standardizing Timestamps and Handling Duplicates:** We converted timestamps to UTC and then to `Europe/Berlin` time. We resolved duplicates caused by the DST "fall-back" transition (where 02:00–03:00 repeats) by retaining the first occurrence of each timestamp, ensuring a unique temporal index.

- **Target Value Cleaning:** Since our goal is supervised regression, we required valid labels for every training instance. We identified and removed 59 rows where the target variables (`count_neckarstadt` or `count_innenstadt`) were missing (`NaN`).

- **Weather Data Imputation:** The weather data provided by the DWD contained gaps. We applied this strategy to handle them:
  1. **Major Gaps:** We defined a day as "unreliable" if it contained more than 5 consecutive hours of missing weather data. We removed these days entirely to prevent the model from learning from defective inputs. This resulted in the removal of 35 days (840 hourly records).
  2. **Minor Gaps:** For smaller, isolated gaps (e.g., 1-2 hours), we used linear interpolation. This assumes that temperature and precipitation change gradually, allowing us to fill the missing values without introducing a lot of noise.

- **Timeframe Restriction:** We restricted the dataset to 01.01.2017 – 31.12.2023. This range excludes partial data from late 2016 and avoids the significant concept drift caused by construction work on the Kurpfalzbrücke in 2024.

3.2 Feature Engineering

We enriched the dataset with external knowledge to capture the temporal and social drivers of bicycle traffic. We generated several new features based on the timestamp and external calendars.

3.2.1 Binning

Bicycle traffic varies significantly depending on the time of day. To help the model capture these non-linear daily patterns, we binned the continuous `hour` attribute into a categorical `time_of_day` feature. We defined the bins as follows:

Time	**Range**	**Time**	**Range**	**Time**	**Range**
Morning	06–10	Midday	10–14	Afternoon	14–18
Evening	18–22	Early Night	22–02	Late Night	02–06

Table 3.1: Binning logic for the time of day feature.

3.3 Data Integration and Feature Management

In the final integration step, we merged the processed bicycle counts with the cleaned weather data and generated features. To define a single target variable representing total bridge throughput, we summed the separate "Neckarstadt" and "Innenstadt" counts into a unified `merged_counts` variable. To reduce dimensionality, redundant attributes such as raw coordinates and the individual side counts were removed following aggregation.

3.3.1 Handling Time Lags

A critical aspect of time-series forecasting is the utilization of past observations (lags) to predict future values. Preliminary analysis suggests that the traffic volume at time $t-1$ is a significantly stronger predictor than static averages. To maintain architectural flexibility, we explicitly excluded hard-coded lag columns from the stored dataset. Since algorithms process history differently (autoregressive models require continuous streams (Section 4.4) while trees rely on distinct feature columns (Section 4.5)) imposing a fixed structure would be restrictive. Furthermore, generating lags dynamically within the training pipeline mitigates data leakage risks during cross-validation. Consequently, the final dataset contains only exogenous features, while temporal lags (e.g., $t-1$, $t-24$) were generated by the specific models themselves.

4 Data Mining

4.1 Hyperparameter Optimization Strategy

We implemented Bayesian Hyperparameter Optimization (HPO) using `Optuna`'s Tree-structured Parzen Estimator (TPE). To respect temporal dependencies, the optimizer was coupled with a Rolling-Origin Cross-Validation scheme (Section 5.1). We employed a **Median Pruner** to halt unpromising trials early, limiting the budget to $N = 20$ trials per model. Analysis of the optimization history revealed a flat loss landscape for polynomial models; the Mean Absolute Error (MAE) remained stable across a broad range of α, indicating robust solutions rather than narrow local minima.

4.2 Linear and Regularized Regression

Linear models provided a baseline for interpretability. We explored standard linear relationships and non-linear interactions via polynomial feature expansion ($d \in \{1, 2, 3\}$). To mitigate overfitting we applied regularization. Numerical features were standardized to zero mean and unit variance using the `StandardScaler`. We compared Ridge Regression (L_2), which shrinks coefficients toward zero, against Lasso Regression (L_1), which performs implicit feature selection by forcing coefficients of redundant predictors to exactly zero. For the Polynomial Ridge model, the optimizer selected a relatively low regularization strength ($\alpha \approx 0.64$), whereas the standard Ridge model landed on a much stronger penalty ($\alpha \approx 9.33$). As mentioned previously in Section 4.1, during HPO the MAE did not change significantly for different α. We observed a similar trend for Lasso regression.

4.3 Lagged Feature Relevance Analysis

We conducted a side experiment to quantify the predictive power of lagged features versus purely exogenous factors (e.g., weather, calendar attributes). Using a Random Forest regressor as a probe, we evaluated five feature strategies.

The study compared a "No-Lag" baseline against cumulative lag configurations. Results indicated that including only the immediate past (`lag_1h`) yields a massive performance jump over no lag at all, outperforming weekly seasonality (`lag_168h`) in isolation. The optimal configuration combines immediate lags (trend), daily lags (circadian rhythm), and weekly lags (seasonality). While this comprehensive temporal context minimizes error, it comes at the compromise of computational efficiency, as the expanded feature space significantly increases the dimensionality and training cost of the model.

4.4 Autoregressive Models

To capture the conditional dependence of demand on past observations, we implemented Autoregressive (AR) models. Rather than relying solely on information criteria, we employed a **Sliding Window Approach** targeting physical cycles:

1. **AR(24) - Daily:** Models the rhythm of the previous day.
2. **AR(168) - Weekly:** Uses the previous values of the last week.
3. **AR(336) - Bi-Weekly:** Extends the historical context to a 14-day horizon.

Forecasting on the test set was performed using a **One-Step-Ahead** approach. The model utilizes ground truth observations from $t-1$ to $t-p$ to predict time t, simulating a real-time production environment where the system updates hourly. This approach proved very computationally efficient.

4.5 Tree-based Ensemble Models

To capture complex, non-monotonic interactions, we evaluated four tree-based ensembles, all tuned using our time series aware validation strategy to prevent data leakage. We first utilized **Random Forest**. The HPO landed on a `max_features` rate of 0.5, indicating that restricting split candidates to 50% of available features was needed to decorrelate the trees and prevent dominant predictors (e.g., `hour_of_day`) from overshadowing subtle signals. We subsequently applied three gradient boosting models, each targeting specific limitations. **XGBoost** was optimized for a "slow learning" strategy, coupling a conservative learning rate ($\eta \approx 0.012$) with high estimators ($N = 565$) and row subsampling to prioritize gradual error reduction over aggressive fitting. **LightGBM** addressed **the extrapolation limits of standard trees** by employing `linear_trees`, which fit piecewise linear regressions within leaf nodes to capture local trends. For LightGBM we scaled the numerical weather data while leaving it unscaled for the other tree ensemble models. Finally, **CatBoost** was selected to handle our categorical features (previously OHE) via Ordered Target Statistics; the optimizer favored a deep tree structure (Depth 9) balanced by strong L_2 leaf regularization to control model complexity.

5 Evaluation

5.1 Time Series Aware Validation

Standard k-fold cross-validation is unsuitable for time-series as it ignores temporal causality. We implemented a **Rolling-Origin Cross-Validation** scheme (Table 5.1), where the training window expands chronologically to predict a future horizon. The final model was evaluated on a held-out test set (Year 2023) using **Mean Absolute Error (MAE)**. Unlike Mean Squared Error (MSE), which squares errors and disproportionately penalizes outliers, MAE provides a linear representation of error in the native unit (bike counts).

5.2 Quantitative Results

Table 5.2 presents the performance on the 2023 test set. The baseline method predicts demand based solely on the historical hourly average.

Table 5.1: Rolling-Origin Cross-Validation Splits

Fold	Training Years	Validation Year	Train Length	Val Length
1	2017–2018	2019	17,325	8,591
2	2017–2019	2020	25,916	8,759
3	2017–2020	2021	34,675	8,591
4	2017–2021	2022	43,266	8,735

5.3 Analysis of Error Distribution

To understand the operational reliability of the best-performing model (CatBoost), we analyzed the residuals ($e_t = y_t - \hat{y}_t$) to identify systematic failure modes.

5.3.1 Conservative Peak Prediction

The model exhibits high precision during low-traffic windows (e.g., 00:00–05:00), where residuals are tightly clustered near zero. However, during the morning rush hour, the model tends to be conservative. While it correctly identifies the onset of the trend, it systematically underestimates the absolute peak volume. This is a known limitation of tree-based ensembles, which average the leaf values and struggle to extrapolate to extreme outliers that lie at the boundaries of the training distribution.

5.3.2 The Holiday Correlation Trap

A specific failure mode was observed during anomalous calendar events, such as New Year's Eve. Our correlation analysis indicates that the `is_holiday` feature generally correlates negatively with bike traffic (due to reduced commuting). The model learns this global rule effectively. However, it fails to account for "festive" exceptions where nightly traffic spikes. Consequently, the model applies the standard "holiday penalty" to New Year's Eve, resulting in a significant underestimation of demand.

5.3.3 Lag Shadowing

In certain high-frequency transition periods, the predictions appear to be "one step behind" the ground truth. This suggests that the model places heavy importance on the 'lag_1h' feature. When exogenous signals (like a sudden weather change) are subtle, the model defaults to mirroring the previous hour's volume. Despite this shadowing effect, the model corrects itself significantly faster than the naive "Last Hour" baseline, confirming that the autoregressive components provide stability but occasionally induce a delay in reaction time.

Table 5.2: Model Performance on Test Set (2023)

Model	MAE (Count)	R^2 Score
Baselines		
Historical Hourly Avg	70.38	0.6756
Last Hour	52.29	0.8309
Regression		
Linear Ridge (No Lags)	77.49	0.6754
Linear Ridge ($\alpha = 9.33$)	39.45	0.8992
Linear Lasso ($\alpha = 0.48$)	39.37	0.8969
Polynomial Ridge ($d = 3, \alpha = 0.64$)	26.70	0.9533
Polynomial Lasso ($d = 3$)	26.80	0.9536
Autoregressive		
AR(24) - Daily	36.00	0.9155
AR(168) - Weekly	28.81	0.9462
AR(336) - Bi-Weekly	26.93	0.9514
Ensemble Trees		
Random Forest	24.45	0.9602
XGBoost	24.53	0.9600
LightGBM (Linear Tree)	24.40	0.9604
CatBoost	24.22	0.9610

5.4 Discussion

5.4.1 Linearity vs. Complexity

The results confirm that urban traffic is inherently non-linear. The Polynomial Ridge model ($d = 3$) significantly outperformed the linear baseline (MAE 39.45 $\rightarrow$ 26.70), proving that interactions between features (like Time and Temperature) are essential.

However, the Ensemble models improved accuracy further (MAE $\approx$ 24.2). While Ensembles are often criticized as "Black Boxes," we argue the trade-off is justified here. A degree-3 polynomial is already too complex for a human to interpret intuitively. Therefore, switching to a high-performance model like CatBoost provides the precision needed for the *BikeReg* traffic signal system without a real loss in "explainability."

5.4.2 The Dominance of Historical Patterns

The hyperparameter search revealed that demand is driven by a broad mix of signals rather than a few specific features. The fact that Lasso (feature selection) performed nearly identically to Ridge (weight shrinking) confirms that the model needs the full spectrum of available data, not just a sparse subset. Crucially, the simple Autoregressive model (AR(336)) performed almost as well as the complex polynomial models (MAE

26.93). This implies that temporal inertia dominates the system: knowing the traffic volume from two weeks ago is often as powerful as knowing the current weather. For deployment, this means that maintaining a reliable historical data pipeline is just as critical as sourcing accurate weather forecasts. Furthermore, we note that due to the specific sensor location and local traffic topology, no direct comparable benchmark exists for the Kurpfalzbrücke for state-of-the-art performance.

5.4.3 The “Lag Shadowing” Risk

CatBoost likely outperformed the other ensembles by handling categorical variables (like ‘Season‘) internally, rather than relying on the sparser One-Hot Encoding used by XGBoost. However, the ”Lag Shadowing” effect observed in our error analysis remains a critical risk. Because the model relies heavily on the previous hour (Lag_1h) to stabilize predictions, it tends to react after a trend has already started. For a ”Green Wave” system, this latency is dangerous. If the model predicts a bicycle platoon only after it has arrived, the traffic light adaptation will be too late. Future iterations must focus on reducing this reaction delay to ensure the system is proactive, not just reactive.

6 Results and Conclusion

This study confirms that dynamic traffic management for the Kurpfalzbrücke is technically feasible. By transitioning from the naive timelag baseline (MAE 52.29) to the optimized CatBoost model (MAE 24.22), we achieved a **53.7% reduction in error**. This demonstrates that the model successfully learned complex patterns beyond simple repetition of the previous hour's volume. The precision effectively supports the proposed “Green Wave” system: during off-peak hours, near-zero residuals ensure the system does not unnecessarily hinder car traffic, while in rush hours, the error is sufficiently low to reliably detect cycling platoons. However, the *lag shadowing* effect identified in our evaluation indicates that the model retains a reactive tendency to sudden trend shifts. To mitigate this latency in a real-world deployment, we suggest investigating a **hybrid control strategy** as a potential solution. In such a setup, the model would serve to define the baseline signal schedule, while real-time data could act as an override trigger for immediate volume spikes. Future research could further explore reducing reaction delay by integrating real-time weather APIs to capture sudden precipitation onset, which our analysis showed to be a immediate driver of traffic reduction. Furthermore, given that the model relies primarily on temporal features and weather data rather than location-specific infrastructure, this framework is likely transferable to other bottlenecks in Mannheim, such as the Jungbuschbrücke. Ultimately, this approach transforms the city's archival data into an actionable forecasting tool, directly supporting Mannheim's goal of becoming a data-driven cycling city.

Bibliography

Centre for Public Impact. Green waves for bicycles in copenhagen. URL https://centreforpublicimpact.org/public-impact-fundamentals/green-waves-for-bicycles-in-copenhagen/. Accessed: 2025-11-30.

Deutscher Wetterdienst. Dwd weather api. URL https://dwd.api.bund.dev. Accessed: 2025-11-29.

Open Holiday API. Public holiday api. URL https://www.openholidaysapi.org/en/. Accessed: 2025-11-29.

Schulferien.org. Schulferien baden-württemberg. URL https://www.schulferien.org/deutschland/ferien/baden-wuerttemberg/. Accessed: 2025-11-29.

Stadt Mannheim. 21-Punkte-Programm Fahrrad. https://www.mannheim.de/de/service-bieten/verkehr/radfahren-in-mannheim/21-punkte-programm, a. Accessed: 2025-11-29.

Stadt Mannheim. Open data portal mannheim, b. URL https://mannheim.opendatasoft.com/page/home/. Accessed: 2025-11-29.

Stadt Mannheim. Dauerradzählstellen für den radverkehr, c. URL https://www.mannheim.de/de/service-bieten/verkehr/radfahren-in-mannheim/dauerradzaehlstellen-fuer-den-radverkehr. Accessed: 2025-11-29.

Ehrenwörtliche Erklärung

Ich versichere, dass ich die beiliegende Bachelor-, Master-, Seminar-, oder Projektarbeit ohne Hilfe Dritter und ohne Benutzung anderer als der angegebenen Quellen und in der untenstehenden Tabelle angegebenen Hilfsmittel angefertigt und die den benutzten Quellen wörtlich oder inhaltlich entnommenen Stellen als solche kenntlich gemacht habe. Diese Arbeit hat in gleicher oder ähnlicher Form noch keiner Prüfungsbehörde vorgelegen. Ich bin mir bewusst, dass eine falsche Erklärung rechtliche Folgen haben wird.

Declaration of Used AI Tools

Tool	Purpose	Where?	Useful?
GPT 5	Latex notation (e.g tables)	Throughout	+
Gemini 3 Pro	Rephrasing and Code Generation	Throughout	+
Gemini 3 Pro	Plot Generation	Throughout	++

Unterschrift

Mannheim, den 30.11.2025